

Local eventual support growth forces aperiodicity for irreducible positive operators on L^p

Autonomous mathematical discovery run `fa_banach_001`
`agent_lane_00` — model GPT5.6

13 August 2026

Abstract

Glück asked whether the local irreducible version of his aperiodicity theorem remains true when one-step support growth is replaced, separately for each positive function, by growth at some pair of consecutive iterates. We give an affirmative answer. If $Tz = \lambda z$ with $|\lambda| = 1$, then $e = |z|$ is a strictly positive fixed vector. Normalizing by e produces a unital positive map R on L^∞ and a unitary $u = z/e$ satisfying $Ru = \lambda u$. The multiplicative-domain identity forces narrow phase slices of u to move into disjoint consecutive phase slices when $\lambda \neq 1$. A nonzero slice inside the prescribed local set therefore violates every possible consecutive support inclusion.

Status. This is a candidate new full proof of the question printed immediately after Theorem 1.2 of [1]. A bounded search found no later resolution; no priority claim is made.

1 The source question and result

Let (Ω, μ) be a σ -finite measure space, let $1 \leq p < \infty$, and let $T : L^p(\Omega, \mu) \rightarrow L^p(\Omega, \mu)$ be positive, power bounded, and irreducible. The source proves that if there is a measurable $S \subseteq \Omega$ of positive measure such that

$$\text{supp}(Tf) \supseteq \text{supp}(f) \quad \text{for every } f \geq 0 \text{ with } \text{supp}(f) \subseteq S,$$

then $\sigma_{\text{pnt}}(T) \cap \mathbb{T} \subseteq \{1\}$. It asks whether the same conclusion follows from the apparently weaker condition

$$\text{for every such } f \text{ there is an } n = n(f) \geq 1 \text{ such that } \text{supp}(T^n f) \supseteq \text{supp}(T^{n-1} f). \quad (1)$$

All support statements are modulo null sets.

Theorem 1. *Under the assumptions above, condition (1) implies*

$$\sigma_{\text{pnt}}(T) \cap \mathbb{T} \subseteq \{1\}.$$

Thus the source question has an affirmative answer, with the integer n allowed to depend on f exactly as in Theorem 1.1 of the source.

2 Idea of the proof

A unimodular eigenvector has a pointwise phase. Irreducibility and power boundedness first force its modulus e to be a strictly positive fixed vector. After dividing by e , the operator becomes a unital positive map R on L^∞ , while the phase $u = z/e$ is a unitary eigenvector of R . Equality in the unital Schwarz inequality then makes u multiplicative for R : a function localized to one phase arc is carried to the corresponding λ -rotated phase arc.

If $\lambda \neq 1$, take an arc narrow enough to be disjoint from its rotation and choose a nonzero piece of that phase arc inside S . All consecutive iterates of this positive piece lie in disjoint rotated arcs. A strictly positive fixed dual functional prevents any iterate from vanishing, so none of the proposed support inclusions can ever occur.

3 Two elementary reductions

We first record why a peripheral eigenvector supplies the order unit needed below. The argument is included because it also guarantees that none of the phase slices can die under iteration.

Lemma 2. *Suppose $Tz = \lambda z$ for a nonzero $z \in L^p$, where $|\lambda| = 1$. Then $e := |z|$ is a fixed vector of T , is strictly positive almost everywhere, and there is a strictly positive functional $\varphi \in (L^p)'_+$ with $T'\varphi = \varphi$.*

Proof. Positivity gives $Te \geq |Tz| = e$. Hence $(T^n e)_{n \geq 0}$ is increasing. It is norm bounded because T is power bounded. The space L^p is a KB-space for $1 \leq p < \infty$, so $T^n e$ converges in norm to some $y \geq e$; continuity gives $Ty = y$.

Choose $\psi \in (L^p)'_+$ with $\langle \psi, y \rangle > 0$. The Cesàro means

$$\varphi_N = \frac{1}{N} \sum_{k=0}^{N-1} (T')^k \psi$$

are norm bounded. A weak-* cluster point φ is positive, $T'\varphi = \varphi$, and $\langle \varphi, y \rangle = \langle \psi, y \rangle > 0$. Its absolute kernel

$$I_\varphi = \{x \in L^p : \langle \varphi, |x| \rangle = 0\}$$

is a closed ideal and is T -invariant, since $|Tx| \leq T|x|$. Irreducibility forces $I_\varphi = \{0\}$, so φ is strictly positive. Finally,

$$0 \leq Te - e, \quad \langle \varphi, Te - e \rangle = 0,$$

and strict positivity imply $Te = e$.

The closed ideal generated by e is nonzero and T -invariant: if $|x| \leq ce$, then $|Tx| \leq cTe = ce$, and the same remains true after taking the closure. Irreducibility therefore makes this ideal all of L^p . Equivalently, $e > 0$ almost everywhere. $\square$

We will also use the following standard multiplicative-domain fact.

Lemma 3 (unitary multiplicative domain). *Let $R : L^\infty(\Omega, \mu) \rightarrow L^\infty(\Omega, \mu)$ be positive and unital. If $u \in L^\infty$ satisfies $|u| = 1$ and $Ru = \lambda u$ for some $|\lambda| = 1$, then, for every $h \in L^\infty$ and every continuous $b : \mathbb{T} \rightarrow \mathbb{C}$,*

$$R(b(u)h) = b(\lambda u)R(h). \tag{2}$$

Proof. A positive map from a commutative C^* -algebra is completely positive. Moreover,

$$R(u^*u) = 1 = (Ru)^*Ru, \quad R(uu^*) = 1 = (Ru)(Ru)^*.$$

Thus u belongs to the multiplicative domain of R . Consequently R is multiplicative on $C^*(u)$ and is a $C^*(u)$ -bimodule map. Approximation of b by trigonometric polynomials gives (2). $\square$

4 Phase-slice obstruction

Proof of Theorem 1. Suppose for a contradiction that $Tz = \lambda z$ for some nonzero z and $\lambda \in \mathbb{T} \setminus \{1\}$. By Lemma 2, $e = |z|$ is fixed and is positive almost everywhere. Put

$$u = \frac{z}{e} \in L^\infty; \quad |u| = 1 \quad \text{almost everywhere.}$$

On the principal order ideal generated by e , define

$$Rg = \frac{T(eg)}{e}, \quad g \in L^\infty. \quad (3)$$

This is well-defined because $|T(eg)| \leq \|g\|_\infty Te = \|g\|_\infty e$. It is a positive unital contraction on L^∞ , and $Ru = \lambda u$.

Because $\lambda \neq 1$, choose a nonnegative continuous function $b : \mathbb{T} \rightarrow [0, 1]$ with nonempty positivity set such that

$$\{b > 0\} \cap \lambda^{-1}\{b > 0\} = \emptyset. \quad (4)$$

We may choose b so that

$$A := S \cap \{b(u) > 0\} \quad \text{has positive measure.} \quad (5)$$

Indeed, cover the circle by finitely many arcs whose closures are disjoint from their λ -rotates, choose continuous bumps positive on smaller arcs which still cover the circle, and use $\mu(S) > 0$.

Let $q = \mathbf{1}_S b(u)$ and $f = eq$. Then $0 \leq f \in L^p$, $\text{supp}(f) \subseteq S$, and (5) together with $e > 0$ almost everywhere gives $f \neq 0$. Repeated use of Lemma 3 yields, for every $n \geq 0$,

$$R^n q = b(\lambda^n u) R^n \mathbf{1}_S, \quad T^n f = e R^n q. \quad (6)$$

It follows from (4) that

$$\text{supp}(T^n f) \cap \text{supp}(T^{n-1} f) = \emptyset \quad (n \geq 1). \quad (7)$$

On the other hand, the strictly positive fixed functional from Lemma 2 gives

$$\langle \varphi, T^n f \rangle = \langle \varphi, f \rangle > 0 \quad (n \geq 0),$$

so every $T^n f$ is nonzero. Hence neither of the two nonzero consecutive supports in (7) can contain the other. This contradicts (1) for the single function f constructed above. Therefore no $\lambda \in \mathbb{T} \setminus \{1\}$ can be an eigenvalue. $\square$

Remark 4. *The proof explains why allowing n to depend on f causes no loss in the irreducible theorem. A nontrivial peripheral eigenvalue creates a measurable phase coordinate u . Once a test function is confined to a sufficiently narrow phase arc, every iterate stays in the corresponding rotated arc, so the obstruction recurs at every time rather than only at time one.*

Verification notes

The only L^p -specific inputs are the KB property for $p < \infty$, the identification of the principal order ideal generated by a strictly positive e with L^∞ via $g \mapsto eg$, and the equivalence between nonvanishing sets and supports modulo null sets. The multiplicative-domain calculation is performed entirely in the commutative C^* -algebra L^∞ . The source question page and a separate audit accompany this packet.

References

- [1] J. Glück, *Aperiodicity of positive operators that increase the support of functions*, arXiv:2209.01171 (2022).